

ADRIANO ZAGAR

Cleveland, OH | (440) 309-9578 | a.b.zagar@vikes.csuohio.edu | www.linkedin.com/in/adrianozagar | adrianozagar.com

EDUCATION

Cleveland State University

B.S. in Electrical and Mechanical Engineering

- GPA: 3.5/4.0

Cleveland, OH

Aug. 2025 – May 2028 (Expected)

EXPERIENCE

Engineering and Manufacturing Intern

Re:Build Manufacturing

Avon, OH

May 2026 – Aug. 2026

- Exceeded production targets by 25%, executing 50+ CNC machining runs
- Increased production rates by 10% by developing an all-new tool inventory tracking system in Microsoft Excel
- Performed 10,000+ inspections using precision measuring tools to maintain strict tolerances
- Mentored incoming interns to accelerate their onboarding and integration into the work environment
- Hands-on experience with advanced CAM software and multi-axis toolpaths

Computer Lab Manager

Cleveland State University

Jan. 2026 – Present

Cleveland, OH

- Managed and maintained 50+ lab computers through hardware diagnostics and OS troubleshooting
- Supported 50+ students and faculty weekly through software, hardware, and network troubleshooting
- Collaborated with IT staff to implement infrastructure improvements and resolve escalated issues

PROJECTS

Team-Developed Rocket and Flight Computer

Embedded C++, ESP32-S3, SOLIDWORKS, OpenRocket

Jun. 2026 – Aug. 2026

- Engineered a custom ESP32-S3 flight computer with sensing, telemetry, and onboard data logging
- Designed and integrated avionics hardware and mechanical components using SOLIDWORKS
- Tested and refined flight hardware iteratively to improve reliability and mission performance

Handheld Embedded Calculator

Arduino IDE, ESP32, KiCad, SOLIDWORKS, C++

May 2026 – Aug. 2026

- Designed and built an ESP32-based calculator integrating custom hardware, C++ firmware, PCB, and enclosure
- Designed a custom 2-layer PCB in KiCad and 3D-printed enclosure in SOLIDWORKS for efficient assembly
- Prototyped and tested breadboard, perfboard, and PCB revisions to improve manufacturability and reliability

Rocketry Club

Rocket Design, Avionics, Embedded Systems, Testing

Jan. 2026 – Present

- Participate in the design, assembly, and testing of student-built rocket systems
- Contribute to avionics and embedded electronics development for flight applications
- Collaborate with team members on design reviews, integration, and launch preparation

Formula SAE

SOLIDWORKS, MATLAB, Vehicle Dynamics, Data Analysis

Nov. 2025 – Present

- Contribute to vehicle assembly and testing activities
- Participate in team design reviews and technical discussions
- Support competition vehicle development efforts

SKILLS

Manufacturing & Machining: CNC Machining, Soldering, Quality Inspection, Assembly, 3D Printing

Automation & Controls: Rockwell PLCs, Allen-Bradley PLCs, FANUC Robotics

CAD & Design: SOLIDWORKS (CSWA), KiCad, EDA, PCB Design

Programming & Software: C++, MATLAB, PSpice, LTspice, Arduino IDE, CAM, OpenRocket, Microsoft Office

Languages: Croatian and some Spanish

CERTIFICATIONS

Certified SOLIDWORKS Associate (CSWA)

Certified Robotics Operator

Fanuc Cert Handling Tool Operations and Programming Certificate

ACTIVITIES & INTERESTS

Organizations: Choose Ohio First / STEM Fellows (2025 – Present)

Clubs: Upper 90 Football Club (2019 – 2025), Men's Soccer/Futsal Clubs (2025 – Present)

Other Interests: Sports, Exercise, Car Maintenance, Electronic Repairs, Video Games